

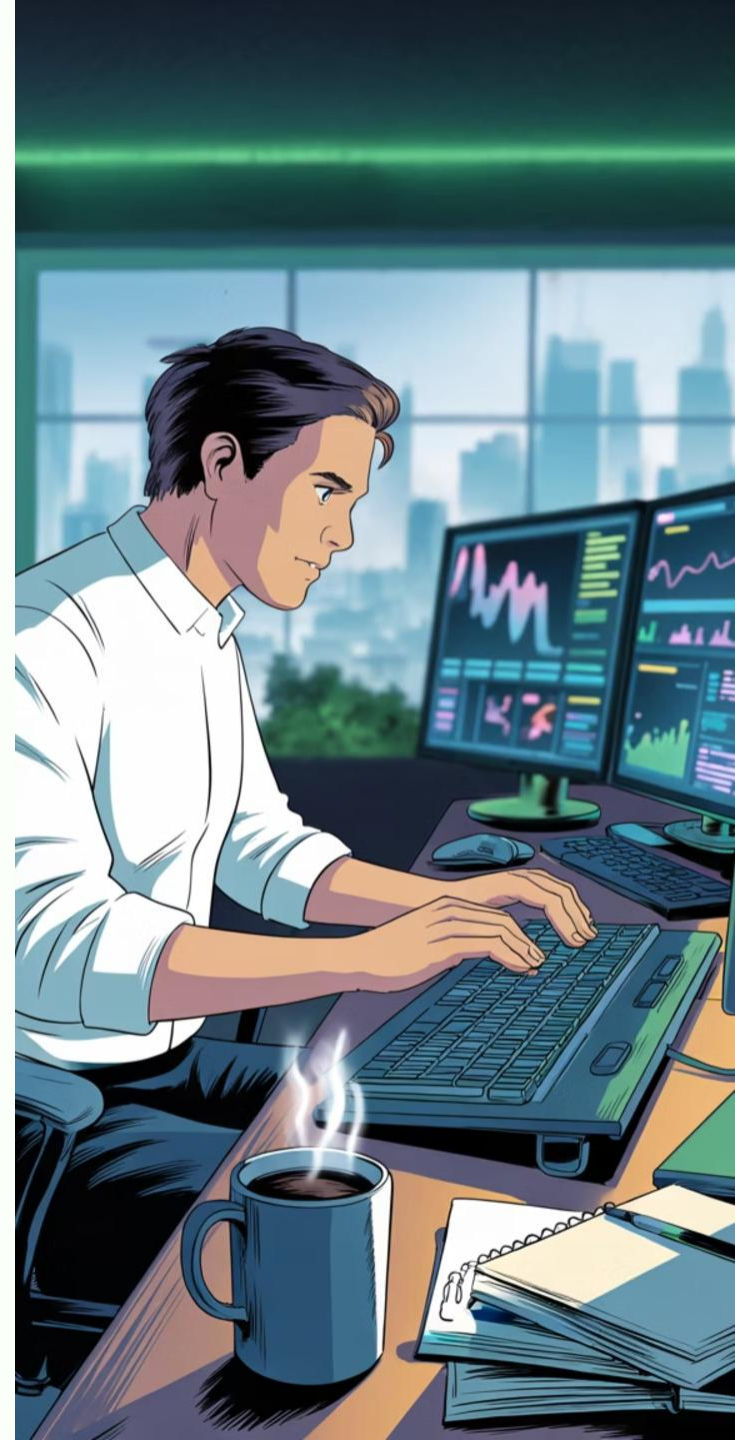
Unit 2 - Part 2

INTRODUCTION TO MACHINE LEARNING USING PYTHON

Unit 2 – Data Preprocessing

Before any Machine Learning model can learn, the data it learns from must be clean, consistent, and meaningful. This unit introduces the foundational concept of **Data Preprocessing** – the essential first step in every Machine Learning pipeline.

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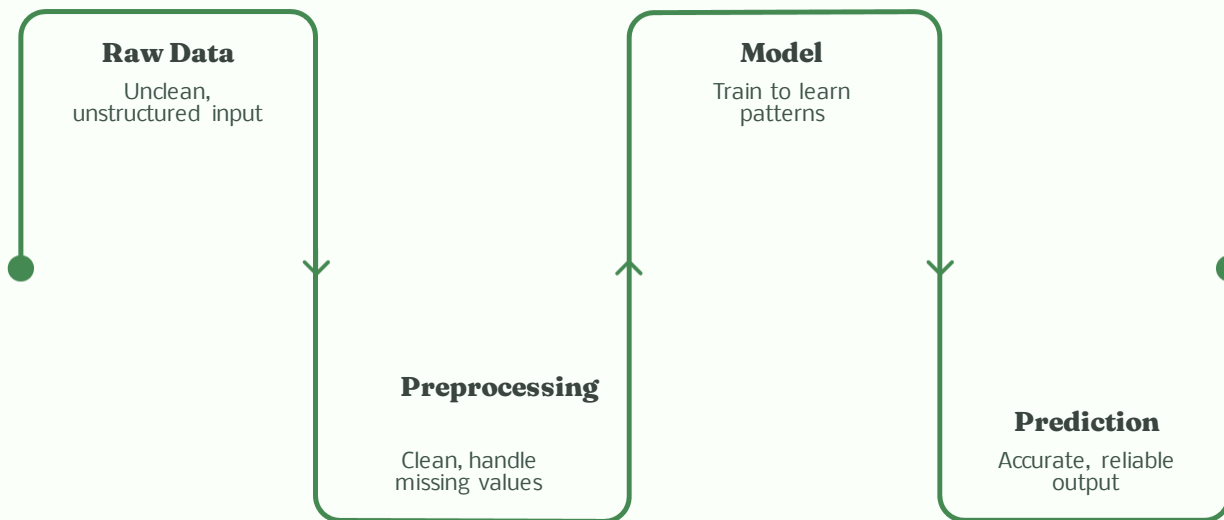


Why Data Preprocessing?

Can we cook vegetables immediately after buying them from the market without washing them?

Machine Learning follows exactly the same principle. Raw data collected from the real world is almost never ready to use directly. It contains errors, gaps, inconsistencies, and irrelevant information – just like unwashed vegetables. Before we can train a Machine Learning model, we must clean and prepare the data. This preparation step is called **Data Preprocessing**.

The Data Preprocessing Pipeline



Each stage in this pipeline depends entirely on the quality of the stage before it. If the data entering the preprocessing stage is poorly structured, even the most sophisticated ML algorithm will struggle to produce meaningful results.

❏ **Garbage In, Garbage Out (GIGO):** This is one of the most important principles in computing and Machine Learning. If you feed poor-quality, dirty, or incorrect data into a model, the predictions it produces will also be poor – no matter how advanced the algorithm. Clean data is not optional; it is the foundation of every successful ML project.

Problems in Raw Data

Real-world datasets are **rarely perfect**. Data collected from surveys, sensors, databases, web scraping, or manual entry almost always contains various types of problems. These problems reduce the accuracy of Machine Learning models and can lead to completely misleading conclusions. Understanding these problems is the first step toward fixing them.

Common Problems Found in Raw Datasets

Problem	Example
Missing Values	Age field is blank (NaN)
Noise	Temperature recorded as 250°C instead of 25°C
Outliers	Salary = ₹25,00,000 while others earn ₹30,000
Duplicate Records	Same student entered twice in the dataset
Incorrect Data	Age = -5 (negative age is impossible)

What This Lecture Covers

This lecture focuses on three of the most critical data quality problems that you will encounter in almost every real-world dataset:

Missing Values

Fields where no value is recorded – represented as **NaN** in Python/Pandas.

Noise

Random or incorrect values that do not represent the true measurement.

Outliers

Data points that are significantly different from the rest of the observations.

Duplicate records and incorrect data entries are also important preprocessing tasks and will be covered during practical lab sessions using Python and Pandas.

Missing Values

What Are Missing Values?

A **missing value** occurs when no data is stored for a particular feature (column) in a specific row of the dataset. In Python and the Pandas library, missing values are represented by the special marker **NaN** – which stands for *Not a Number*.

Consider this student dataset:

Student	Age
Amit	22
Priya	25
Rahul	NaN
Sneha	30

Rahul's age is missing – it shows as **NaN**. This is a missing value.

Why Are Missing Values harmful?

Incorrect Analysis

Statistical summaries like averages and totals become inaccurate when missing values are not handled properly.

Reduced Accuracy

Models trained on incomplete data learn incomplete patterns, resulting in poor predictions.

Why Do Missing Values Occur?

- **Human Error:** Data entry operators forget to fill certain fields, skip questions on forms, or make typing mistakes during manual entry.
- **Machine or Sensor Failure:** Automated data collection systems may malfunction – a broken sensor fails to record a temperature reading, or a network interruption causes data loss during transfer.
- **Privacy Concerns:** Respondents may intentionally leave sensitive fields blank – income, medical history, or personal identifiers are commonly omitted.
- **Data Loss During Transfer:** When data moves between systems, databases, or file formats, some values may be lost or corrupted in the process.

Wrong Averages

Many tools either ignore or misinterpret NaN values, leading to skewed calculations.

Algorithm Errors

Many ML algorithms (like Linear Regression, SVM) cannot process NaN values at all and will crash.

Methods to Handle Missing Values

There is no single "correct" way to handle missing values. The best method depends on the **type of data**, the **amount of missing data**, and the **context of the problem**. Below are the four most commonly used techniques.

1

Delete the Row

Simply remove the entire row (record) that contains the missing value. This is the quickest approach and works well when only a very small number of rows have missing data. However, if many rows are deleted, you lose valuable information and your dataset shrinks significantly.

Best for: Very few missing values; large datasets where losing a few rows does not matter.

2

Replace with Mean

Calculate the average (mean) of all available values in that column and use it to fill the missing entries. This preserves the dataset size and works well for numerical data that is roughly symmetrically distributed without extreme outliers.

Best for: Numerical data with a normal distribution and no significant outliers.

3

Replace with Median

Use the middle value (median) of the column to fill missing entries. The median is less affected by extreme values than the mean, making it a more robust choice when the data is skewed or contains outliers.

Best for: Numerical data that is skewed or contains outliers (e.g., salary, house prices).

4

Replace with Mode

Use the most frequently occurring value (mode) in the column. This is the most appropriate method for **categorical data** (non-numerical data like gender, city, product category) where calculating a mean or median does not make sense.

Best for: Categorical data (e.g., filling a missing "City" field with the most common city).

Quick Comparison Table

Method	Best Used For	Advantage	Disadvantage
Delete Row	Few missing values	Simple and quick	Loss of information
Mean	Numerical, symmetric data	Preserves dataset size	Skewed by outliers
Median	Numerical data with outliers	Robust to extremes	May not represent all cases
Mode	Categorical data	Works for non-numeric fields	May over-represent one category

Noise in Data

What Is Noise?

Noise refers to unwanted, random, or incorrect information in a dataset that does not represent the true or intended value. Noise is essentially "dirty data" – values that are present but wrong. Unlike missing values (where data is absent), noisy data is present but misleading.

Consider these real examples:

Correct Value	Noisy Value
Temperature = 25°C	Temperature = 250°C
Salary = ₹50,000	Salary = ₹99,99,999
Age = 20	Age = 500
Attendance = 85%	Attendance = 850%

These noisy values are physically impossible or highly improbable. If a Machine Learning model learns from them, it will form incorrect patterns.

Why Is Noise Harmful?

Noise **confuses Machine Learning algorithms**. When a model sees a temperature of 250°C or an age of 500 years, it tries to learn patterns from these impossible values. This leads to:

- Inaccurate predictions on real-world data
- Models that do not generalise well
- Wasted computational resources learning from garbage

Techniques for Reducing Noise

Correct obvious errors – manually or programmatically fix values that are clearly wrong (e.g., negative ages).

Remove incorrect records – if a value cannot be corrected reliably, remove that record entirely.

Causes of Noisy Data



Typing Mistakes

Manual data entry errors – extra zeros, transposed digits, or wrong units entered by the operator.



Measurement Errors

Incorrect calibration of instruments leads to systematically wrong measurements across many records.



Sensor Malfunction

Faulty hardware produces incorrect readings – a broken thermometer might output 250°C instead of 25°C.



Data Transmission Errors

During data transfer between systems, values can get corrupted, duplicated, or misformatted.

Outliers

What Is an Outlier?

An **outlier** is a data point that is **significantly different** from the other observations in a dataset. It lies far away from the majority of the data points and does not follow the general pattern.

Consider this salary dataset for employees in a company:

Employee	Monthly Salary
Employee A	₹30,000
Employee B	₹31,000
Employee C	₹29,000
Employee D	₹32,000
Employee E	₹35,000
Employee F (CEO)	₹25,00,000

The average salary of the first five employees is approximately **₹31,400**. But when we include the CEO's salary, the average jumps to over **₹4,44,500** – a completely misleading figure for most employees. The CEO's salary of ₹25,00,000 is an **outlier**.

Real-Life Examples of Outliers

Human Age = 250 Years

Biologically impossible – clearly a data entry error or sensor malfunction.

Weight = 500 kg

Far beyond the normal human weight range – likely a unit error (grams vs kg).

Electricity Bill = ₹15

When the typical bill is ₹1,500 – possibly a decimal error or missing digits.

Large Banking Transaction

A single ₹50,00,000 transfer among hundreds of small ₹500-₹2,000 transactions – may indicate fraud.

❏ **Important:** Not every outlier is an error! Some outliers represent genuine but rare events – a billionaire's income, a record-breaking temperature, or a viral social media post. Always investigate the context before deciding to remove an outlier.

Detecting Outliers

Before we can handle outliers, we must first **identify** them. Several techniques are commonly used in data analysis and Machine Learning to detect unusual values in a dataset. In this lecture, we cover the concepts – mathematical formulas will be explored in practical sessions.



Box Plot

A **Box Plot** (or Box-and-Whisker Plot) is a graphical representation that displays the distribution of data. It shows the median, quartiles, and any values that fall outside the "whiskers" – these outer points are flagged as potential outliers. Box plots are one of the most popular and intuitive visual tools for outlier detection.



Z-Score (Concept)

The **Z-Score** measures how many standard deviations a data point is away from the mean (average). A Z-Score of 0 means the value equals the mean. Values with a Z-Score greater than +3 or less than -3 are typically considered outliers, as they lie more than 3 standard deviations from the average.



Scatter Plot

A **Scatter Plot** displays individual data points on a two-dimensional graph. Outliers appear as isolated points that are far away from the main cluster of observations. This method is especially useful when analysing the relationship between two numerical variables and spotting unusual combinations.



Interquartile Range (IQR)

The **IQR** is the range between the first quartile (Q1, 25th percentile) and the third quartile (Q3, 75th percentile) – representing the middle 50% of the data. Any value below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ is flagged as a potential outlier. This is one of the most robust methods as it is not affected by extreme values.

Note for Students: You are not required to memorise mathematical formulas for Z-Score or IQR in this lecture. Focus on understanding the *concepts* – what each method does and when to use it. Implementation details will be covered in Python practical sessions using libraries like Matplotlib, Seaborn, and Pandas.

Summary and Classroom Activity

Key Takeaways

01

Raw data cannot be used directly for Machine Learning – it must be preprocessed first.

02

Data preprocessing improves data quality and directly improves model accuracy.

03

Garbage In, Garbage Out (GIGO) – poor-quality input data always leads to poor-quality predictions.

04

Missing values occur due to incomplete information and can be handled using Delete, Mean, Median, or Mode.

05

Noise represents incorrect or random values that confuse ML algorithms and must be cleaned.

06

Outliers are observations that differ significantly from the rest – detect them using Box Plots, Scatter Plots, Z-Score, or IQR.

Classroom Activity

Identify the data quality problem in each example below. Discuss with your classmates and write your answer.

Example	Identify the Problem
Age = NaN	?
Salary = ₹99,99,999	?
Temperature = 250°C	?
One employee earns ₹25,00,000 while everyone else earns ₹30,000	?

"A Machine Learning model is only as good as the data it learns from. Clean data is the foundation of every successful Machine Learning project."